

HOMEWORK 4, DUE OCTOBER 1

(10 points) 1. Let $(x_n)_n$ and $(y_n)_n$ be two sequences of real numbers, so that $(y_n)_n$ is increasing. Assume that

$$\lim_{n \rightarrow \infty} \frac{x_{n+1} - x_n}{y_{n+1} - y_n} = L,$$

for some $L \in \mathbb{R}$. Prove that for any $\varepsilon > 0$ there exists $N > 0$ so that

$$(L - \varepsilon)(y_{n+1} - y_N) < x_{n+1} - x_N < (L + \varepsilon)(y_{n+1} - y_N),$$

for all $n > N$.

Hint: Use $x_{n+1} - x_N = (x_{n+1} - x_n) + (x_n - x_{n-1}) + \dots + (x_{N+1} - x_N)$, for any $n > N$.

(10 points) 2. Let $(x_n)_n$ and $(y_n)_n$ be two sequences of real numbers, so that $(y_n)_n$ is increasing and $\lim_{n \rightarrow \infty} y_n = \infty$. Assume that

$$\lim_{n \rightarrow \infty} \frac{x_{n+1} - x_n}{y_{n+1} - y_n} = L,$$

for some $L \in \mathbb{R}$. Prove that

$$\lim_{n \rightarrow \infty} \frac{x_n}{y_n}$$

exists and is also equal to L .

Hint: Use problem 1.

(10 points) 3. Prove that if

$$\lim_{n \rightarrow \infty} a_n = L,$$

then

$$\lim_{n \rightarrow \infty} \frac{a_1 + a_2 + \dots + a_n}{n}$$

exists and equals L .

(10 points) 4. Define the sequence $a_1 = 1$ and

$$a_{n+1} = 3 - \frac{1}{a_n}.$$

Show that a_n is convergent, by proving that a_n is monotone and bounded. What is the limit of a_n ?